About the Team / Organisation

BetaRevive is building Gluco, a Self Evolving healthcare operations agent.

Rai Biswas, team leader: led ideation and shaped the product direction.
Dr. Manikanda Krishnan, doctor partner: helped validate the idea.
Vishruth (wysh), technical lead: built the application, defined its technical scope and drove improvements.

Problem Statement (use case specific)

During DKA/HHS care, a recorded order can still await collection, a collected specimen can lack lab receipt, and a validated result can await acknowledgement. Staff must find the gap and the person responsible across departments and shifts.

Initial users: doctors, nurses, laboratory staff and care coordinators. The proposed benefit is less manual chasing, to be measured in a hospital pilot.

AI Solution Description

Gluco combines chat and visual workflows with a Self Evolving loop: capture confirmed outcomes as Care Routines, test them, obtain approval and reuse them with another encounter's evidence. Today, this is bounded operational learning under human control.

Technology and Tools Used

React/TypeScript, OpenUI, Fastify and PostgreSQL. Gluco uses a customized Hermes-based runtime and Sarvam speech. Vercel hosts the web app and Azure runs the backend. HIS/LIS events are simulated.

Key Features of the Solution

Chat with visual evidence and care pipelines
Role-specific work and shift-wide monitoring
Edited handovers with named receiving staff
Tested, approved and versioned Care Routines
Source-confirmed closure and audit history

Build Sprint Progress

Expanded report review into shared inpatient coordination. Added staff roles, visual pipelines, shift monitoring and edited handovers. Simplified navigation and voice controls. Fixed event-reconciliation races and delayed guide opening through regression tests.

User Validation and AI Evaluation

7 Nov: 420 automated tests passed. Browser coverage: 204 distinct passing cases across runs, with two desktop-only cases skipped on mobile. Hosted checks passed monitoring, handovers, authorization and approved routine reuse.

These are fictional-data engineering tests, not real staff feedback or clinical validation. Latest speech tests mocked provider responses. Live-model performance and a hospital pilot remain to be validated.

Safety and Responsible AI

Fictional data only. No diagnosis, prescribing or clinical decision support. Server-side permissions and human approval govern actions. Routine reuse checks new evidence. Missing evidence remains unknown.

Deployment and Sustainability Plan

Proposed 60–90 day pilot in one ward. Agree sources and owners, measure a baseline, then run shadow mode. Compare receipt and acknowledgement delays, chasing time and alert burden. A facility subscription is a hypothesis to validate.

Supporting Links

MVP: https://glucoflow.vercel.app/
Select Explore the clinic. No registration or manual credentials required.
Code: https://github.com/wysh3/glucoflow